Connor Day

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Education

University of California, San Diego — San Diego, CA

Bachelor of Science in Data Science — *Expected June 2028*

- **Relevant Coursework:** Exploratory Data Analysis and Inference, Data Management (SQL), Systems for Scalable Analytics, Recommender Systems and Web Mining, Probability, Data Visualization, Practice and Application of Data Science, Theoretical Foundations of Data Science I & II

Projects

Garment Search | *Python, PyTorch, Hugging Face Transformers, FAISS, Pandas, NumPy* — Sep. 2026 – Oct. 2026

- Built a text-to-image search engine over 45,622 Fashionpedia clothing photos, comparing two vision-language embedding models, fashion-specific FashionCLIP and general-purpose CLIP, via exact FAISS indexes
- Pre-registered the evaluation before embedding any images: 20 queries across four difficulty tiers, written relevance criteria, pass thresholds, and six predicted failure modes
- FashionCLIP outperformed CLIP on capped recall@10 (0.612 vs. 0.488) and hand-judged precision@10 (0.755 vs. 0.540), but both failed the compositional tier and neither returned a clearly oversized blazer (0 of 10)
- Hand-judged 391 query-image pairs blind to model and rank, finding that 28% of results the dataset's labels scored as irrelevant were clearly relevant, so label-based recall understated retrieval quality

Thyroid and Metabolic Health | *Python, Pandas, Polars, statsmodels, scikit-learn, Matplotlib* — Aug. 2026

- Pre-registered four falsifiable hypotheses linking thyroid function to metabolic health, then tested them on three pooled NHANES cycles (2007–2012) covering 3,459 U.S. adults
- Fit survey-weighted regression models using NHANES strata, PSUs, and sample weights, adjusting for age, sex, and BMI, and built a metabolic syndrome flag from the 2009 Harmonized clinical criteria
- Applied FDR and Bonferroni correction: three of four hypotheses were unsupported, and the one significant result ran opposite to the prediction (+14.9 mg/dL triglycerides per unit FT3/FT4 ratio, $p < 0.001$)
- Built a reproducible pipeline from raw NHANES survey files to per-hypothesis analytic samples

Power Outages and Population Density | *Python, Pandas, scikit-learn, Plotly* — Course Project, June 2026

- Tested whether population density predicts outage duration across 1,534 major U.S. outages (2000–2016); a 10,000-iteration permutation test found no significant difference ($p = 0.065$)
- Built a scikit-learn Random Forest pipeline tuned with 5-fold GridSearchCV on features known at outage start, cutting test RMSE 12% below a mean-only baseline (6,888 vs. 7,839 minutes)
- Used permutation tests to analyze missingness in customer counts and check model fairness across density groups

Experience

Polo Ralph Lauren Factory Store — Livermore, CA

Front & Back of House Team Member — *June 2022 – Sep. 2025*

- Selected by management to collaborate with top-performing team members on styling store sections for corporate lookbook photography, adopted company-wide as the seasonal floor standard for all Ralph Lauren factory stores
- Presented floor sets and store walks to store management and visiting corporate leadership, explaining the reasoning behind layout and merchandising decisions
- Trained 10+ new team members on store procedures, POS operations, and customer service standards, explaining processes clearly to people with no prior retail experience

Technical Skills

Languages: Python, SQL (proficient); Java, R, C++ (coursework)
Libraries: Pandas, NumPy, Polars, statsmodels, scikit-learn, PyTorch, Hugging Face, FAISS, Plotly, Matplotlib
Data Skills: Model benchmarking, predictive modeling, hypothesis testing, survey-weighted regression, FDR correction
Tools: Git, GitHub, Jupyter, LaTeX, Claude Code, Codex